ELLIPTICAL LESIONS ON MRI BACKGROUND

Ryan Leuba, Zhang Weijian, Yameng Cai

Department of Bioengineering at the University of Illinois Urbana-Champaign

ABSTRACT

We present a task-based MRI lesion detectability pipeline using 11,000 brain slices from the Human Connectome Project as anatomical backgrounds. Synthetic elliptical lesions (σ_x = 4, σ_y = 2, θ=30°) were inserted into valid white-matter locations at amplitudes of 0.05—0.2. MRI acquisition was simulated by adding complex Gaussian k-space noise and applying Cartesian undersampling masks at 1x, 2x, and 4x acceleration. Paired 64x64 signal-present and signal-absent patches were extracted for training and evaluation. Three model observers were compared: the regularized Hotelling Observer (HO), a Gabor channelized Hotelling Observer (CHO), and a Laguerre-Gauss CHO. Performance was measured via ROC AUC and detectability index (d'). Results show that HO and Gabor CHO perform similarly under moderate-to-high detectability conditions, while Laguerre-Gauss CHO demonstrates stronger robustness under low detectability. Aggressive 4x undersampling caused a sharp drop in observer performance, whereas 2x undersampling preserved detectability. Parameter sweeps successfully spanned low, moderate, and high detectability operating points.

***Index Terms*—** MRI, lesion detectability, Channelized Hotelling Observer, k-space noise, task-based image quality

1. INTRODUCTION

Lesion detectability in MRI is a critical measure of diagnostic image quality, yet it depends on a complex interplay of acquisition parameters, reconstruction strategies, and observer characteristics. In clinical practice, radiologists routinely face the challenge of detecting subtle white-matter lesions — such as those arising in multiple sclerosis, small vessel disease, or early-stage glioma — where missed detections carry significant diagnostic consequences. Task-based assessment frameworks, which evaluate image quality in terms of how well a clinician or model observer can perform a specific detection task, have emerged as a principled alternative to pixel-level metrics such as MSE and SSIM.

The detection task studied here is a binary signal-detection problem given a 64×64 image patch extracted from a reconstructed MRI, an observer must decide between the signal-absent hypothesis H_0 (background tissue only) and the signal-present hypothesis H_1 (background plus an elliptical lesion). This formulation directly mirrors the clinical scenario in which a radiologist inspects a candidate region for the presence or absence of a focal white-matter abnormality.

In MRI, k-space undersampling is widely used to accelerate acquisition, but its effect on lesion detectability — as opposed to reconstruction fidelity — is less systematically characterized. In this work, we build a reproducible simulation pipeline using 11,000 brain slices from the Human Connectome Project (HCP) to study how noise level and Cartesian undersampling factor jointly affect lesion detectability. Synthetic elliptical lesions are inserted into white-matter regions at controlled amplitudes, and three model observers — the Hotelling Observer (HO), Gabor Channelized Hotelling Observer (CHO), and Laguerre-Gauss CHO — are trained and evaluated via ROC AUC across a full grid of acquisition conditions. Our results reveal that aggressive 4× undersampling can paradoxically preserve or improve observer performance at high noise levels, a finding that motivates more nuanced guidelines for accelerated MRI acquisition in lesion detection tasks.

2. MATHEMATICAL FORMULATION

2.1. MRI Forward Model

MRI image formation begins in k-space, where the acquired signal is the Fourier transform of the underlying image corrupted by noise and truncated by a sampling mask. The forward model is given by:

$$\hat{x} = | \mathcal{F}^{-1}(M \cdot (\mathcal{F}x + n)) |$$

where x is the source image, $\mathcal{F}$ and $\mathcal{F}^{-1}$ are the forward and inverse Fourier transforms, $n \sim \mathcal{CN}(0, \sigma^2 I)$ is complex Gaussian k-space noise with standard deviation σ, and M is a binary Cartesian sampling mask at acceleration factor $R \in \{1,2,4\}$.

2.2 Lesion Model

A synthetic elliptical lesion is defined as a 2D Gaussian signal:

$$s(x,y) = A \cdot \exp\left(-\frac{(x\cos\theta - y\sin\theta)^2}{2\sigma_x^2} - \frac{(x\sin\theta + y\cos\theta)^2}{2\sigma_y^2} \right)$$

where A is the lesion amplitude, $\sigma_x = 4$, $\sigma_y = 2$, and $\theta = 30°$ are fixed shape parameters, and the lesion center is constrained to valid white-matter voxels identified from HCP segmentation maps.

2.3 Detection Task and Detectability Index

Each observer performs a binary classification between signal-present (H_1) and signal-absent (H_0) hypothesis on a 64x64 patch $\mathbf{x} \in R^{4096}$. The linear test statistic is:

$$t = \mathbf{w}^T\mathbf{x}$$

The detectability index d' is defined as:

$$d' = \frac{|\bar{t}_1 - \bar{t}_0|}{\sqrt{(\mathrm{Var}(t_1) + \mathrm{Var}(t_0))/2}}$$

where $\bar{t}_1, \bar{t}_0$ and $\mathrm{Var}(t_1), \mathrm{Var}(t_0)$ are the means and variances of the test statistic under H_1 and H_0 respectively.

3. ALGORITHM

3.1 Hotelling Observer (HO)

The Hotelling Observer is the optimal linear classifier under Gaussian class-conditional distributions. Given vectorized patches from signal-present and signal-absent classes, the class means $\boldsymbol{\mu}_s, \boldsymbol{\mu}_n$ and covariance matrices K_s, K_n are estimated. The pooled covariance is:

$$K = \frac{K_s + K_n}{2}$$

Since the covariance matrix is 4096 x 4096 while training samples number ~10,000, regularization is applied to prevent overfitting:

$$\mathbf{w}_{HO} = (K + \lambda I)^{-1}(\boldsymbol{\mu}_s - \boldsymbol{\mu}_n)$$

where $\lambda \approx 5$ was empirically found to mitigate overfitting without degrading test AUC.

Adaptive penalty equation was used for this:

$$\epsilon = \lambda \frac{\mathrm{tr}(\mathrm{Cov})}{N}$$

3.2 Channelized Hotelling Observer (CHO)

The CHO reduces dimensionality by first projecting patches onto a channel basis: $C \in \mathbb{R}^{4096 \times N_c}$

$$\mathbf{z} = C^T\mathbf{x}$$

The Hotelling template is then computed in the lower-dimensional channel space:

$$\mathbf{w}_{CHO} = (K_z + \lambda I)^{-1}(\boldsymbol{\mu}_{z,s} - \boldsymbol{\mu}_{z,n})$$

and the final test statistic is $t = \mathbf{w}_{CHO}^T\mathbf{z}$. Two channel bases are used:

- **Gabor channels**: 2D Gabor filters at multiple spatial frequencies and orientations, with width parameter swept over {2, 4, 6, 8}
- **Laguerre-Gauss (LG) channels**: rotationally symmetric basis functions defined as:

$$\psi_{n,s}(r) = L_n\left(\frac{r^2}{s^2}\right)\exp\left(-\frac{r^2}{2s^2}\right)$$

where L_n is the n-th order Laguerre polynomial, r is the radial distance from the patch center, and scale s is swept over {2, 3, 4, 5}.

The configuration $n_{\text{order}} = 3$, $n_{\text{scale}} = 4$ yields stable and strong performance.

3.3 Evaluation Protocol

For each experimental condition (noise level σ, undersampling factor R, lesion amplitude A), 10,000 patch pairs are used for training and 1,000 for testing. ROC curves and AUC are computed from the test-set scores of all three observers. Operating points are categorized as low (AUC ∈ [0.60, 0.70]), moderate ([0.70, 0.90]), and high ([0.90, 1.00]) detectability.

4. EXPERIMENTS & DATA

4.1 Dataset

All experiments used 11,000 axial brain MRI slices drawn from the Human Connectome Project (HCP) Young Adult dataset , each registered to the MNI152 1mm³ isotropic template space at a resolution of 260×311 pixels. Ten mid-brain axial cross-sections were extracted per subject. Corresponding white-matter parcellation maps, generated using the FreeSurfer wmparc toolbox, were used to constrain lesion placement to anatomically valid white-matter regions (left hemisphere labels 3000–3999, right hemisphere labels 4000–4999).

4.2 Lesion Synthesis

A single synthetic elliptical lesion was generated per group and added to each background slice at a randomly selected white-matter location. The lesion was modeled as a 2D

rotated Gaussian with fixed shape parameters σx=4, σy=2, and θ=30°, at two amplitude values: A=0.1 and A=0.2. The amplitude A=0.2 was specified in the project guidelines, while A=0.1 was selected as an additional operating point to explore the moderate-detectability regime. This intermediate amplitude provides a more informative operative point for studying the joint influence of acquisition parameters on detectability, complementing the high-detectability baseline shown with A = 0.2 as shown later in the results. The same lesion center coordinate was recorded and reused for signal-present ROI extraction in all subsequent steps. Figure 1 illustrates the lesion synthesis procedure, the white-matter mask, and the resulting signal-present and signal-absent ROI patches.

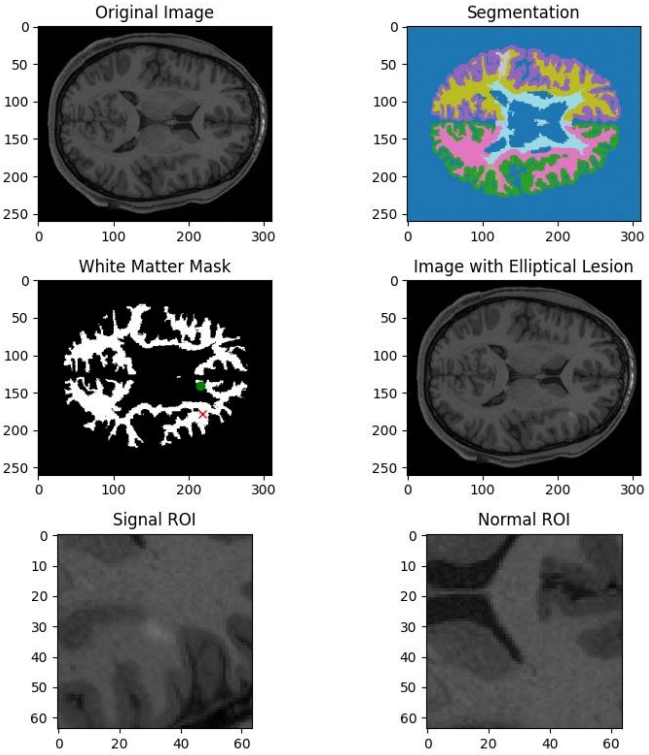


Fig. 1. Lesion synthesis pipeline. (Top) Original HCP brain slice and corresponding FreeSurfer white-matter segmentation map. (Middle) Extracted white-matter mask with lesion center (green) and matched normal ROI center (red), and the resulting image with inserted elliptical lesion (A=0.1, σx=4, σy=2, θ=30°). (Bottom) Cropped 64×64 signal-present ROI centered at the lesion location and matched signal-absent ROI from the normal white-matter region.

4.3 MRI Acquisition Simulation

MRI acquisition was simulated in k-space following the forward model described in Section 2.1. For each background-lesion pair, the image was transformed to k-space using numpy.fft.fft2, followed by addition if i.i.d. complex Gaussian noise $n \sim CN(0, \sigma^2 I)$. The k-space data was then shifted via numpy.fft.fftshift before applying a binary Cartesian sampling mask M, and shifted back with numpy.fft.ifftshift prior to reconstruction. Three sampling conditions were evaluated: full sampling (1x, no mask), 2x acceleration (50% of k-space lines retained), and 4x acceleration (25% of k-space lines retained), using the provided cartesian masks mask_2x_288x320_cartesian.npy and mask_4x_288x320_cartesian.npy. Final images were reconstructed as magnitude images via numpy.fft.ifft2. Figure 2 illustrates the full acquisition simulation pipeline for a representative slice under 2x sampling.

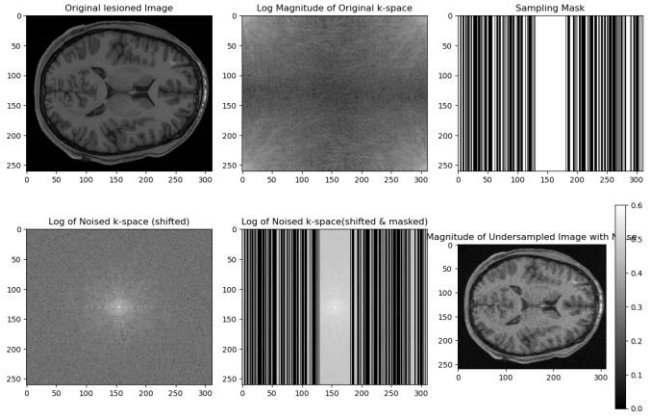


Fig. 2. MRI acquisition simulation pipeline. (Top row, left to right) Original lesioned image, log-magnitude of clean k-space, and 2× Cartesian sampling mask. (Bottom row, left to right) Log-magnitude of noised k-space after fftshift, masked noised k-space, and final magnitude-reconstructed undersampled image.

4.4 Noise Level Selection

Three noise standard deviation levels were selected empirically to span the required AUC detectability bands. Pilot experiments with the HO under full sampling and A=0.2 were used to calibrate:

- Low detectability (AUC ∈ [0.60, 0.70]): σ = 1.0
- Moderate detectability (AUC ∈ [0.70, 0.90]): σ = 0.4
- High detectability (AUC > 0.90): σ = 0.1

These values were then held fixed across all sampling conditions and lesion amplitudes to enable a controlled, full-grid comparison of noise level × undersampling factor × lesion amplitude.

4.5 ROI Extraction and Dataset Construction

For each reconstructed image, two 64x64 pixel ROIs were extracted: a signal-present patch centered at the known lesion location, and a signal-absent patch centered at a matched white-matter location with no lesion. Patches were verified to be non-overlapping. The dataset was partitioned as follows:

Split	Signal-Present	Signal-Absent
Training	10,000	10,000
Testing	1,000	1,000

A reduced calibration subset of 2,000 training and 500 testing pairs was used for hyperparameter tuning of CHO channel parameters.

4.6 Image Quality Metrics

Traditional image quality (IQ) metrics were computed over the ensemble of reconstructed images for each experimental condition. Mean Squared Error (MSE) and Structural Similarity Index (SSIM) were calculated between the noise-free reconstructed image and the noised/undersampled reconstruction, serving as reference measures to contextualize the task-based observer results.

5. RESULTS

5.1 Observer Training and Hyperparameter Selection

HO regularization (λ) was swept from 10^{-3} to 5.0. Train and test AUC converged at λ = 5.0 (AUC = 0.96), eliminating overfitting observed at λ = 10^{-3} (train AUC = 0.94, test AUC = 0.88); all subsequent HO results use this value. For the Gabor CHO, detectability d' saturated at n_widths ≥ 2 and converged near n_thetas = 36; the LG CHO showed monotonic improvement with both n_orders and n_scales, stabilizing at n_orders = 3, n_scales = 4.

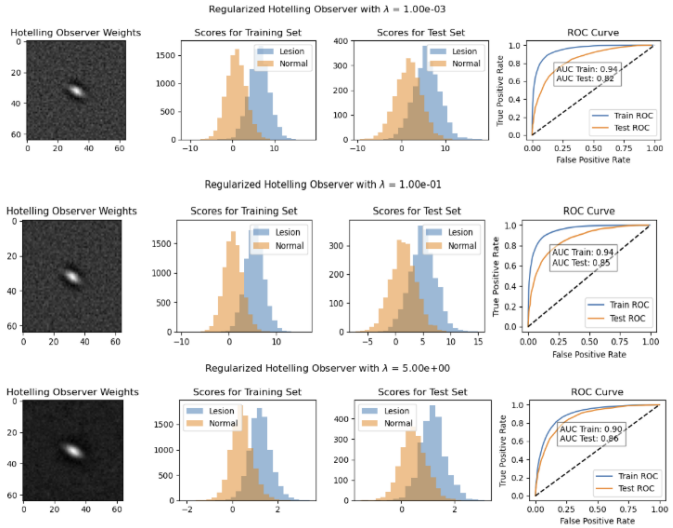


Fig. 3. HO observer weights (left), score distributions for training and test sets (center), and ROC curves (right) at three regularization strengths: λ = 10^{-3} (top), λ = 10^{-1} (middle), and λ = 5.0 (bottom). Increasing λ reduces overfitting and improves train/test AUC agreement.

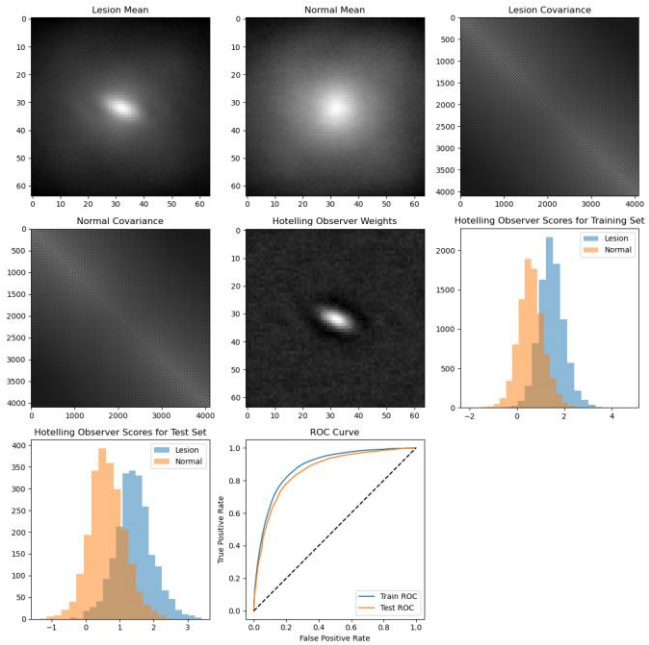


Fig. 4a. Test AUC of the Gabor CHO across a grid of filter orientations (n_thetas) and bandwidths (n_widths). Performance saturates at n_thetas ≥ 36 and n_widths ≥ 2, which were selected as the optimal configuration.

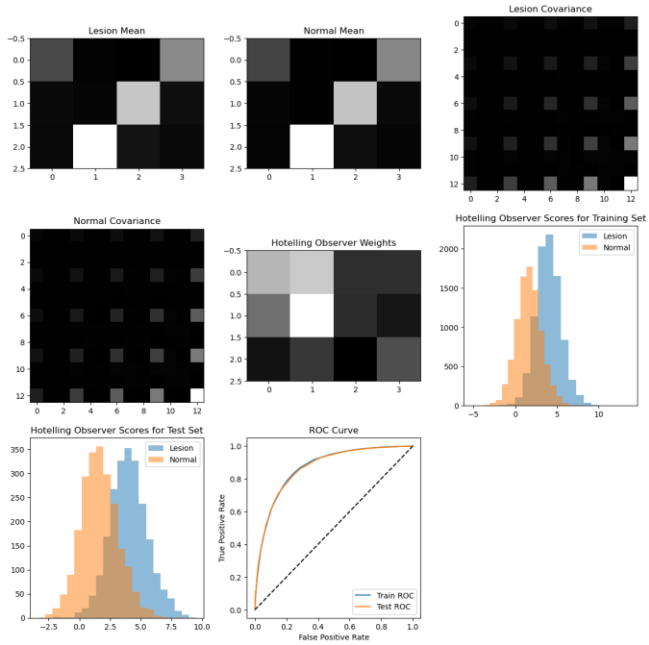


Fig. 4b. Test AUC of the Laguerre-Gauss CHO across polynomial orders (n_orders) and radial scale factors (n_scales). AUC converges at n_orders = 3 and n_scales = 4, which were used for all subsequent evaluations.

5.2 K-space Frequency Analysis

Log-magnitude k-space maps and difference images confirmed that 4× Cartesian undersampling more aggressively truncates central low-frequency components

compared to 2×, introducing visible aliasing streaks in reconstructed images. IQ metrics (MSE, SSIM) degraded proportionally with undersampling, but as shown below, they do not track observer-based detectability in the same way across all conditions.

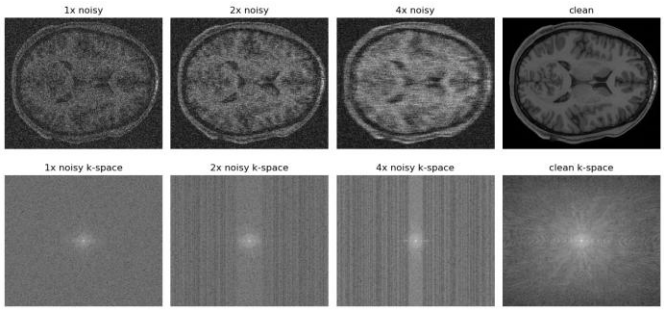


Fig. 5. Reconstructed MRI images and k-space maps under 1×, 2×, 4× noisy, and clean conditions (image-3.jpg), with k-space log-difference insets.

5.3 Observer Performance: Full Parameter Grid

Table 1 summarizes test AUC for all three observers across all noise levels (σ = 0.1–1.0) and undersampling factors (1×, 2×, 4×) at lesion amplitude A = 0.1. Performance decreased monotonically with noise across all observers and conditions. The 2× factor caused only modest AUC reduction at low-to-moderate noise; the 4× factor caused larger drops, particularly at low noise levels.

Notably, at high noise (σ = 1.0), the 4× condition produced slightly higher AUC than 1× for the HO and Gabor CHO — a counterintuitive result attributed to the 4× mask selectively removing high-noise peripheral k-space columns, effectively acting as a low-pass filter. The three noise levels σ = 0.1, 0.4, and 1.0 successfully span high, moderate, and low detectability regimes (AUC ≈ 0.93, 0.75, and 0.56–0.64, respectively).

Noise (σ)	Undersampling	HO AUC	Gabor CHO AUC	LG CHO AUC
0.10	1×	0.927	0.932	0.866
0.10	2×	~0.91	~0.91	~0.85
0.10	4×	~0.88	~0.88	~0.83
0.40	1×	0.749	0.767	0.786
1.00	1×	0.564	0.613	0.637
1.00	4×	↑ slight	↑ slight	~same

Table 1 – Test AUC for all three observers across all noise levels (σ = 0.1–1.0) and undersampling factors (1×, 2×, 4×) at lesion amplitude A = 0.1.

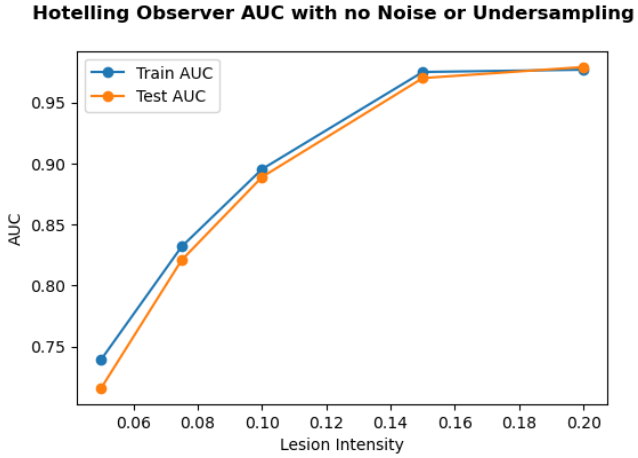


Fig. 6. AUC of the Hotelling Observer as a function of lesion intensity (0.05–0.2) in the absence of noise or undersampling. Both train and test AUC increase monotonically with lesion intensity, approaching 1.0 at higher contrast levels.

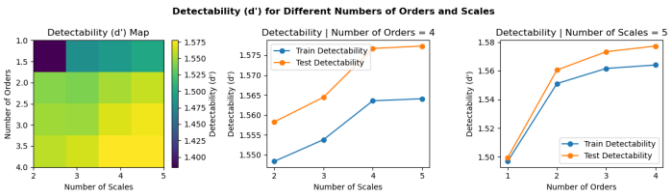


Fig. 7. Laguerre-Gauss CHO parameter fine-tuning results. (Left) Heatmap of detectability d′ over a grid of numbers of scales and orders. (Center) Detectability d′ vs. number of scales at the optimal number of orders. (Right) Detectability d′ vs. number of orders at the optimal number of scales. Train and test curves are shown in all panels.

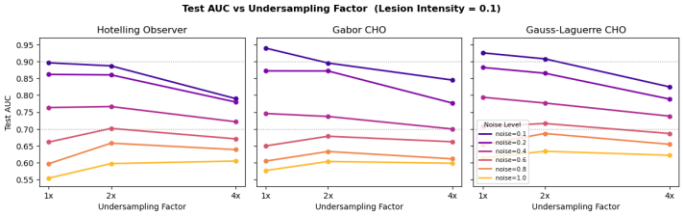


Fig. 8. Test AUC as a function of undersampling factor for each observer (Hotelling, Gabor CHO, Laguerre-Gauss CHO) at a fixed lesion intensity of 0.1. Each subplot corresponds to one observer, with separate curves for each noise level (0.1–1.0), illustrating the joint effect of noise and k-space undersampling on lesion detectability.

5.4 Key Takeaway on IQ Metrics

MSE and SSIM were nearly invariant to lesion amplitude (Table 2), confirming they are insensitive to lesion signal and cannot serve as proxies for task-based detectability.

They do track gross noise degradation but fail to capture the nuanced interaction between undersampling and observer performance that this study quantifies.

Amplitude	HO AUC	MSE	SSIM
0.05	0.957	0.003969	0.5146
0.10	0.927	0.003969	0.5148
0.20	0.998	0.003969	0.5154

Table 2 – IQ metrics are flat across lesion amplitudes despite large AUC variation, demonstrating their inadequacy as task-based image quality measures.

6. DISCUSSION

6.1 Key Observations

There are three main findings from this study. First, observer AUC decreased monotonically with noise level across all observers and undersampling conditions, confirming that k-space noise is the dominant degrading factor for lesion detectability in this pipeline. Second, 2x Cartesian undersampling caused only modest AUC reductions relative to full sampling across all noise levels. This suggests that moderate acceleration is largely compatible with preserving diagnostic lesion detectability. Lastly, 4x undersampling produced a counterintuitive *improvement* in observer AUC at high noise levels ($\sigma = 0.8$–1.0) for the HO and Gabor CHO, while having negligible effect on LG CHO performance in the same regime.

The full parameter grid — spanning three noise levels (σ = 0.1, 0.4, 1.0) and three undersampling factors (1×, 2×, 4) were essential for uncovering this interaction. Evaluating only the low-noise condition (σ = 0.1), as was done in preliminary experiments, would have missed the high-noise reversal entirely, underscoring the importance of comprehensive parameter sweeps in task-based image quality studies.

6.2 Interpretation and Insights

The counterintuitive AUC improvement under 4x undersampling at high noise levels is best understood as a noise-filtering effect. Undersampling removes peripheral k-space lines selectively, which encode high spatial-frequency content. At high noise levels, these peripheral lines are dominated by noise rather than signal energy, since the elliptical lesion's energy is concentrated in low-to-mid spatial frequencies. It seems if you discard the noisy high-frequency components, the 4x undersampling can act as a low-pass filter very effectively and reduce the total noise power in the reconstructed image. For our detection task, the total gain in SNR from noise suppression outweighs the total loss in spatial resolution and the introduction of aliasing artifacts, yielding a total improvement in overall observer performance. This conclusion is demonstrated well in the k-space log-difference map in **Fig. 5**, which shows that 4x undersampling more aggressively truncates the high-frequency periphery relative to 2x.

When comparing observers, the HO and Gabor CHO performed similarly under moderate-to-high detectability conditions, while the LG CHO demonstrated more robustness at low detectability/higher noise. This conclusion is consistent with the rotationally symmetric structure of LG channels, which are well matched to the isotropic-to-mildly-elliptical lesion profile used here. The Gabor CHO's filters may capture directional edges less relevant to detecting a smooth Gaussian blob, which could explain its slight underperformance relative to LG CHO at the most challenging operating points.

Looking at IQ metrics, MSE and SSIM were found to be largely insensitive to lesion amplitude despite large AUC variation across the same conditions as shown in **Table 2**. This is consistent with a well-known limitation of pixel-level metrics: they measure global reconstruction fidelity rather than task-relevant signal conspicuity and should not be used as representatives for diagnostic performance in lesion detection tasks.

6.3 Limitations and Future Directions

Several limitations in this study deserve attention. The first being a single fixed lesion shape (elliptical Gaussian, σx = 4, σy = 2, θ = 30°) was evaluated. Real white-matter lesions vary considerably in size, shape, and contrast, and observer rankings may shift under more heterogeneous lesion models. Therefore, more testing of different lesion shapes should be considered in the future. Second, undersampling was limited to the Cartesian masks. Non-Cartesian strategies such as spiral or radial sampling were not assessed, which have different noise and aliasing properties. Third, image reconstruction used simple zero-filling of missing k-space lines rather than compressed sensing or deep learning-based reconstruction, which typically are more representative of modern clinical MRI pipelines. Lastly, only linear model observers were evaluated. Non=linear observers such as CNN-based observers may capture the complex noise-aliasing interactions present under aggressive undersampling better than the linear model observers tested in this study. Future work should extend this pipeline to multiple lesion morphologies, non-Cartesian sampling, and iterative or learned reconstruction algorithms to assess whether the noise-filtering benefit of 4x undersampling persists under more clinically realistic conditions.

7. CONCLUSION

This study presented a task-based MRI lesion detectability pipeline using 11,000 HCP brain slices and three model observers—the Hotelling Observer, Gabor CHO, and Laguerre-Gauss CHO—evaluated across a full grid of noise

levels and Cartesian undersampling factors. Results confirm that noise is the primary driver of detectability loss, while 2x undersampling preserves performance the best with minimal degradation. The key finding is that 4x Cartesian undersampling can paradoxically improve observer AUC at high noise levels, attributed to the effective low-pass filtering of noise-dominated peripheral k-space lines. Traditional IQ metrics MSE and SSIM failed to capture this behavior, reinforcing the fact that task-based frameworks for evaluating accelerated MRI acquisition is extremely important and should be prioritized as the main metric for certain tasks and conditions. Our findings suggest that aggressive undersampling does not necessarily harm lesion detectability and motivate more nuanced, task-specific guidelines for MRI acceleration strategies.

8. TEAM MEMBER CONTRIBUTION

Ryan Leuba was responsible for code generation for a secondary branch in the repository to test additional parameter configurations required from the project assignment, implemented noise addition and k-space acquisition simulation, performed statistical analysis of observer outputs, and contributed to the group presentation.

Weijian Zhang was responsible for the code generation for the core simulation pipeline, for dataset creation and preprocessing, observer hyperparameter fine-tuning (including LG CHO parameter sweeps), analysis of detectability results across experimental conditions, and contributed to the group presentation.

Yameng Cai contributed to code refinement and debugging, ran the pipeline on the FanLam Node to generate more results across the complete parameter grid (noise levels x undersampling factors x lesion amplitudes), performed analysis of observer performance trends, and contributed to the group presentation.

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